The degree and extremal number of edges in hamiltonian connected graphs

Tung-Yang Ho*
Department of Information Management
Ta Hwa Institute of Technology, Hsinchu, Taiwan 30740, R.O.C.

Cheng-Kuan Lin and Jimmy J. M. Tan
Department of Computer Science
National Chiao Tung University, Hsinchu, Taiwan 30010, R.O.C.

D. Frank Hsu
Department of Computer and Information Science
Fordham University, New York, NY 10023, U.S.A.

Lih-Hsing Hsu
Department of Computer Science and Information Engineering
Providence University, Taichung, Taiwan 43301, R.O.C.

Abstract

Assume that n and δ are positive integers with $3 \leq \delta < n$. Let $hc(n, \delta)$ be the minimum number of edges required to guarantee an n-vertex graph G with $\delta(G) \geq \delta$ to be hamiltonian connected. Any n-vertex graph G with $\delta(G) \geq \delta$ is hamiltonian connected if $|E(G)| \geq hc(n, \delta)$. We prove that $hc(n, \delta) = C(n - \delta + 1, 2) + \delta^2 - \delta + 1$ if $\delta \leq \lfloor \frac{n+3\times(n \bmod 2)}{6} \rfloor + 1$, $hc(n, \delta) = C(n - \lfloor \frac{n}{2} \rfloor + 1, 2) + \lfloor \frac{n}{2} \rfloor^2 - \lfloor \frac{n}{2} \rfloor + 1$ if $\lfloor \frac{n+3\times(n \bmod 2)}{6} \rfloor + 1 < \delta \leq \lfloor \frac{n}{2} \rfloor$, and $hc(n, \delta) = \lceil \frac{n\delta}{2} \rceil$ if $\delta > \lfloor \frac{n}{2} \rfloor$.

1 Introduction

For the graph definitions and notations, we follow [1]. Let $G = (V, E)$ be a graph if V is a finite set and E is a subset of $\{(u, v) \mid (u, v)$ is an unordered pair of $V\}$. We say that V is the *vertex set* and E is the *edge set*. Two vertices u and v are *adjacent* if $(u, v) \in E$. The *complete graph* K_n is the graph with n vertices such that any two distinct vertices are adjacent. The *degree* of a vertex u in G, denoted by $\deg_G(u)$, is the number of vertices adjacent to u. We use $\delta(G)$ to denote $\min\{\deg_G(u) \mid u \in V(G)\}$. A *path* of length $m-1$, $\langle v_0, v_1, \ldots, v_{m-1} \rangle$, is an ordered list of distinct vertices such that v_i and v_{i+1} are adjacent for $0 \leq i \leq m-2$. A *cycle* is a path with at least three vertices such that the first vertex is the same as the last one. A *hamiltonian cycle* of G is a cycle that traverses every vertex of G exactly once. A graph is *hamiltonian* if it has a hamiltonian cycle. A *hamiltonian path* is a path of length $|V(G)| - 1$. A graph G is *hamiltonian connected* if there exists a hamiltonian path between any two distinct vertices of G. It is easy to see that a hamiltonian connected graph with at least three vertices is hamiltonian.

It is proved by Moon [9] that the degree of any vertex in a hamiltonian connected graph with at least four vertices is at least 3, so it is natural to consider the n-vertex graph G with $n \geq 4$ and $\delta(G) \geq 3$. Assume that n and δ are positive integers with $3 \leq \delta < n$. Let $hc(n, \delta)$ be the minimum number of edges required to guarantee an n-vertex graph with $\delta(G) \geq \delta$ to be hamiltonian connected. Any n-vertex graph G with $\delta(G) \geq \delta$ is hamiltonian connected if $|E(G)| \geq hc(n, \delta)$. We will prove the following main theorem.

*Correspondence to: hoho@thit.edu.tw

Theorem 1 *Assume that n and δ are positive integers with $3 \le \delta < n$. Then $hc(n,\delta) = C(n-\delta+1,2)+\delta^2-\delta+1$ if $\delta \le \lfloor \frac{n+3\times(n \bmod 2)}{6} \rfloor + 1$, $hc(n,\delta) = C(n-\lfloor \frac{n}{2} \rfloor+1,2)+\lfloor \frac{n}{2} \rfloor^2-\lfloor \frac{n}{2} \rfloor+1$ if $\lfloor \frac{n+3\times(n \bmod 2)}{6} \rfloor + 1 < \delta \le \lfloor \frac{n}{2} \rfloor$, and $hc(n,\delta) = \lceil \frac{n\delta}{2} \rceil$ if $\delta > \lfloor \frac{n}{2} \rfloor$.*

We defer the proof of Theorem 1. In Section 2, we present the mathematical background. Finally, we give the proof of Theorem 1 in Section 3.

2 Preliminary

The following theorem is proved by Ore [10].

Theorem 2 *[10] Let G be an n-vertex graph with $\delta(G) > \lfloor \frac{n}{2} \rfloor$. Then G is hamiltonian connected.*

The following theorem is given by Lick [8].

Theorem 3 *[8] Let G be an n-vertex graph. Assume that the degree d_i of G satisfy $d_1 \le d_2 \le \ldots \le d_n$. If $d_{j-1} \le j \le n/2 \Rightarrow d_{n-j} \ge n-j+1$, then G is hamiltonian connected.*

To our knowledge, no one has ever discussed the sharpness of the above theorem. In the following, we give a logically equivalent theorem.

Theorem 4 *Let G be an n-vertex graph. Assume that the degree d_i of G satisfy $d_1 \le d_2 \le \ldots \le d_n$. If G is non-hamiltonian connected, then there exist at least one integer $2 \le m \le n/2$ such that $d_{m-1} \le m \le n/2$ and $d_{n-m} \le n-m$.*

To discuss the sharpness of Theorem 4, we introduce the following family of graphs. Let $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ be two graphs. The *union* of G_1 and G_2, written $G_1 + G_2$, has edge set $E_1 \cup E_2$ and vertex set $V_1 \cup V_2$ with $V_1 \cap V_2 = \emptyset$. The *join* of G_1 and G_2, written $G_1 \vee G_2$, obtained from $G_1 + G_2$ by joining each vertex of G_1 to each vertex of G_2.

The *degree sequence* of an n-vertex graph is the list of vertices degree, in nondecreasing order, as $d_1 \le d_2 \le \ldots \le d_n$. For $2 \le m \le n/2$, let $H_{m,n}$ denote the graph $(\bar{K}_{m-1} + K_{n-2m+1}) \vee K_m$. The graphs $H_{3,11}$ and $H_{4,12}$ are shown in Figure 1. Obviously, the degree sequence of $H_{m,n}$ is $(\underbrace{m, m, \ldots, m}_{m-1}, \underbrace{n-m, n-m, \ldots, n-m}_{n-2m+1}, \underbrace{n-1, n-1, \ldots, n-1}_{m})$.

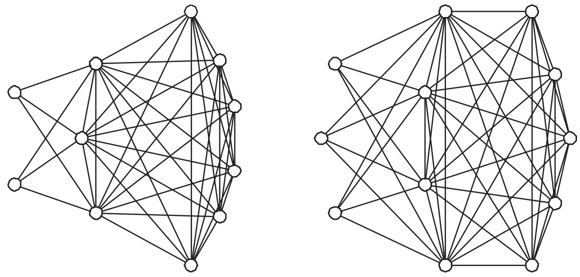

Figure 1: The graphs (a) $H_{3,11}$ and (b) $H_{4,12}$

A sequence of real numbers $(p_1, p_2, \ldots, p_n)$ is said to be *majorised* by another sequence $(q_1, q_2, \ldots, q_n)$ if $p_i \le q_i$ for $1 \le i \le n$. A graph G is *degree-majorised* by a graph H if $|V(G)| = |V(H)|$ and the nondecreasing degree sequence of G is majorised by that of H. For instance, the 5-cycle is degree majorised by the complete bipartite graph $K_{2,3}$ because $(2,2,2,2,2)$ is majorised by $(2,2,2,3,3)$.

Lemma 1 *Let $G = (V, E)$ be a graph, X be a subset of V, and u, v be any two distinct vertices in X. Suppose that there exists a hamiltonian path between u and v. Then there are at most $|X|-1$ connected components of $G - X$.*

Let S be the subset of $V(H_{m,n})$ corresponding to the vertex of K_m. Since $2 \le m \le n/2$, $|S| \ge 2$. Let u and v be any two distinct vertices in S. Obviously, there are m connected components of $H_{m,n} - S$. By Lemma 1, $H_{m,n}$ does not have a hamiltonian path between u and v. Thus, $H_{m,n}$ is not hamiltonian connected. In other words, the result in Theorem 4 is sharp.

So we have the following corollary.

Corollary 1 *The graph $H_{m,n}$ is not hamiltonian connected where n and m are integers with $2 \le m \le n/2$.*

Thus, the following theorem is equivalent to Theorem 4.

Theorem 5 *If G is an n-vertex non-hamiltonian connected graph, then G is degree-majorised by some $H_{m,n}$ with $2 \le m \le n/2$.*

Corollary 2 *Let $n \ge 6$. Assume that G is an n-vertex non-hamiltonian connected graph. Then*

$\delta(G) \leq \lfloor \frac{n}{2} \rfloor$ *and* $|E(G)| \leq \max\{|E(H_{\delta(G),n})|, |E(H_{\lfloor \frac{n}{2} \rfloor,n})|\}$.

Proof. Let G be any n-vertex non-hamiltonian connected graph. With Theorem 2, $\delta(G) \leq \lfloor \frac{n}{2} \rfloor$. By Theorem 5, G is degree-majorised by some $H_{m,n}$. Since $\delta(H_{m,n}) = m$, $\delta(G) \leq m \leq \lfloor \frac{n}{2} \rfloor$. Therefore $|E(G)| \leq \max\{|E(H_{m,n})| \mid \delta(G) \leq m \leq \lfloor \frac{n}{2} \rfloor\}$. Since $|E(H_{m,n})| = \frac{1}{2}(m(m-1)+(n-2m+1)(n-m)+m(n-1))$ is a quadratics function with respect to m and the maximum value of it occurs at the boundary $m = \delta(G)$ or $m = \lfloor \frac{n}{2} \rfloor$, $|E(G)| \leq \max\{|E(H_{\delta(G),n})|, |E(H_{\lfloor \frac{n}{2} \rfloor,n})|\}$. $\diamondsuit$

By Corollary 2, we have the following corollary.

Corollary 3 *Let G be an n-vertex graph with $n \geq 6$. If $|E(G)| \geq \max\{|E(H_{\delta(G),n})|, |E(H_{\lfloor \frac{n}{2} \rfloor,n})|\} + 1$, then G is hamiltonian connected.*

Lemma 2 *Let n and k be integers with $n \geq 6$ and $3 \leq k \leq \lfloor \frac{n}{2} \rfloor$. Then $|E(H_{k,n})| \geq |E(H_{\lfloor \frac{n}{2} \rfloor,n})|$ if and only if $3 \leq k \leq \lfloor \frac{n+3\times(n \bmod 2)}{6} \rfloor + 1$ or $k = \lfloor \frac{n}{2} \rfloor$.*

Proof. We first prove the case that n is even. We claim that $|E(H_{k,n})| \geq |E(H_{\frac{n}{2},n})|$ if and only if $3 \leq k \leq \lfloor \frac{n}{6} \rfloor + 1$ or $k = \frac{n}{2}$. Suppose that $|E(H_{k,n})| < |E(H_{\frac{n}{2},n})|$. Then $|E(H_{k,n})| = \frac{1}{2}(k(k-1)+(n-2k+1)(n-k)+k(n-1)) < |E(H_{\frac{n}{2},n})| = \frac{1}{2}((\frac{n}{2}-1)(\frac{n}{2}))+(\frac{n}{2})(n-1)+(\frac{n}{2})$. This implies $3k^2-(2n+3)k+(\frac{1}{4}n^2+\frac{3}{2}n) < 0$, which means $(k-\frac{n}{2})(3k-\frac{n}{2}-3) < 0$. Thus $|E(H_{k,n})| < |E(H_{\frac{n}{2},n})|$ if and only if $\frac{n}{6}+1 < k < \frac{n}{2}$. Note that n and k are integers with n is even, $n \geq 6$, and $3 \leq k \leq \frac{n}{2}$. Therefore, $|E(H_{k,n})| \geq |E(H_{\frac{n}{2},n})|$ if and only if $3 \leq k \leq \lfloor \frac{n}{6} \rfloor + 1$ or $k = \frac{n}{2}$.

For odd integer n, using the same method, we can prove that $|E(H_{k,n})| < |E(H_{\frac{n-1}{2},n})|$ if and only if $\frac{n+3}{6}+1 < k < \frac{n-1}{2}$. Given that $n \geq 7$, and $3 \leq k \leq \frac{n-1}{2}$, then $|E(H_{k,n})| \geq |E(H_{\frac{n-1}{2},n})|$ if and only if $3 \leq k \leq \lfloor \frac{n+3}{6} \rfloor + 1$ or $k = \frac{n-1}{2}$. Therefore, the result follows. $\diamondsuit$

3 Proof of Theorem 1

Now, we give the proof of Theorem 1.

By brute force, we can check that $hc(4,3) = 6$, $hc(5,3) = 8$, and $hc(5,4) = 10$. Therefore, the theorem holds for $n = 4, 5$.

Then, we consider that $3 \leq \delta \leq \lfloor \frac{n}{2} \rfloor$ and $n \geq 6$.

Suppose that $3 \leq \delta \leq \lfloor \frac{n+3\times(n \bmod 2)}{6} \rfloor + 1$ or $\delta = \lfloor \frac{n}{2} \rfloor$. By Lemma 2, $|E(H_{\delta,n})| \geq |E(H_{\lfloor \frac{n}{2} \rfloor,n})|$. Let G be any n-vertex graph with $\delta(G) \geq \delta$ and $|E(G)| \geq |E(H_{\delta,n})| + 1$. By Corollary 3, G is hamiltonian connected. By the definition of $H_{m,n}$. We know that $|E(H_{\delta,n})| + 1 = C(n-\delta+1, 2) + \delta^2 - \delta + 1$. Therefore, $hc(n,\delta) \leq C(n-\delta+1,2)+\delta^2-\delta+1$. By Corollary 1, $H_{\delta,n}$ is not hamiltonian connected. Thus, $hc(n,\delta) > |E(H_{\delta,n})| = C(n-\delta+1,2)+\delta^2-\delta$. Hence, $hc(n,\delta) = C(n-\delta+1,2)+\delta^2-\delta+1$.

Suppose that $\lfloor \frac{n+3\times(n \bmod 2)}{6} \rfloor + 1 < \delta < \lfloor \frac{n}{2} \rfloor$. By Lemma 2, $|E(H_{\delta,n})| < |E(H_{\lfloor \frac{n}{2} \rfloor,n})|$. Let G be any n-vertex graph with $\delta(G) \geq \delta$ and $|E(G)| \geq |E(H_{\lfloor \frac{n}{2} \rfloor,n})|+1$. By Corollary 3, G is hamiltonian connected. We note that $|E(H_{\lfloor \frac{n}{2} \rfloor,n})|+1 = C(n-\lfloor \frac{n}{2} \rfloor+1,2)+\lfloor \frac{n}{2} \rfloor^2-\lfloor \frac{n}{2} \rfloor+1$. Therefore, $hc(n,\delta) \leq C(n-\lfloor \frac{n}{2} \rfloor+1,2)+\lfloor \frac{n}{2} \rfloor^2-\lfloor \frac{n}{2} \rfloor+1$. By Corollary 1, $H_{\lfloor \frac{n}{2} \rfloor,n}$ is not hamiltonian connected. Thus, $hc(n,\delta) > |E(H_{\lfloor \frac{n}{2} \rfloor,n})| = C(n-\lfloor \frac{n}{2} \rfloor+1,2)+\lfloor \frac{n}{2} \rfloor^2-\lfloor \frac{n}{2} \rfloor$. Hence, $hc(n,\delta) = C(n-\lfloor \frac{n}{2} \rfloor+1,2)+\lfloor \frac{n}{2} \rfloor^2-\lfloor \frac{n}{2} \rfloor+1$.

Finally, we consider the case that $\delta > \lfloor \frac{n}{2} \rfloor$ and $n \geq 6$. Let G be any graph with $\delta(G) \geq \delta > \lfloor \frac{n}{2} \rfloor$. By Theorem 2, G is hamiltonian connected. Obviously, $|E(G)| \geq \lceil \frac{n\delta}{2} \rceil$. Thus, $hc(n,\delta) = \lceil \frac{n\delta}{2} \rceil$.

The proof is complete.

References

[1] J. A. Bondy, and U. S. R. Murty, "Graph Theory with Applications," North-Holland, New York, 1980.

[2] M. Y. Chan and S. J. Lee, "On the existence of Hamiltonian circuits in faulty hypercubes," SIAM Journal on Discrete Mathematics, 4, p.511–527, 1991.

[3] J. S. Fu, "Conditional fault hamiltonicity of the complete graph," Information Processing Letters, 107, p.110–113, 2008.

[4] T. Y. Ho, Y. K. Shih, J. J. M. Tan, and L. H. Hsu, "Conditional fault hamiltonian connecticity of the complete graph," to appear in Information Processing Letters, 2009.

[5] W. T. Huang, J. J. M. Tan, C. N. Hung, and L. H. Hsu, "Fault-tolerant hamiltonicity of twisted cubes," Journal of Parallel and Distributed Computing, 62, p.591–604, 2002.

[6] S. Latifi, S. Q. Zheng, and N. Bagherzadeh, "Optimal ring embedding in hypercubes with faulty links," in Proceedings of the IEEE Symposium on Fault-Tolerant Computing, p.178–184, 1992.

[7] T. K. Li, J. J. M. Tan, and L. H. Hsu, "Hyper hamiltonian laceability on the edge fault star graph," Information Sciences, 165, p.59–71, 2004.

[8] D. R. Lick, "A sufficient condition for hamiltonian connectedness," Journal of Combinatorial Theory, 8, p.444-445, 1970.

[9] J. W. Moon, "On a problem of Ore," Math. Gaz., 49, p.40–41,1965

[10] O. Ore, "Hamilton connected graphs," Journal of Mathematic Pures Application, 42, p.21–27, 1963.

[11] O. Ore, "Note on Hamilton circuits," The American Mathematical Monthly, 67, p.55, 1960.